

Adnan Ahmed Khan

AI Engineer | Applied NLP, Knowledge Graphs & Production Systems

Islamabad, Pakistan

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SUMMARY

AI and backend engineer with 6+ years shipping Python and Node.js systems. Built a multimodal content-integrity pipeline that generates video descriptions, compares them with user-provided text, and flags suspected semantic mismatches. Developed Graphiti/Neo4j temporal memory, hybrid graph/vector retrieval, and semantic post-condition verification for LangGraph agents. Production owner for platforms serving 50k–100k DAU and 500–800 RPS.

TECHNICAL SKILLS

Applied AI/NLP: LLM applications, multimodal video understanding, semantic consistency and drift detection, structured outputs, embeddings, prompt engineering

Knowledge Systems: Neo4j, Graphiti, temporal knowledge graphs, graph modeling, hybrid graph/vector retrieval, Chroma, Elasticsearch

Backend/Data: Python, TypeScript, JavaScript, FastAPI, Node.js, Nest.js, REST, PostgreSQL, Redis, Kafka, MongoDB, ETL

Cloud/Operations: AWS, Kubernetes, Docker, Terraform, Helm, GitHub Actions, ArgoCD, Prometheus, Grafana, Jaeger, distributed systems

PROFESSIONAL EXPERIENCE

HackOnTech

2023 – Present

Senior Software Engineer, Backend & Platform

Islamabad / Remote

- Built a multimodal NLP workflow that generates descriptions from video, compares them with user-supplied descriptions, and flags suspected semantic mismatches for review.
- Led a zero-to-one overhaul into six Docker/EKS microservices, enabling zero-downtime releases for products serving 50k–100k DAU.
- Stabilized 500–800 RPS bursts and cut core PostgreSQL/Redis query latency from 30s+ to under 1s through indexing, caching, pooling, and service decomposition.
- Reduced failed cross-service operations by 80% and MTTR from approximately 45 minutes to under 10 minutes using Temporal recovery, idempotency, tracing, and service-level dashboards.
- Cut AWS spend by 25% through EKS rightsizing and autoscaling while preserving service objectives.

Stack: Python, FastAPI, TypeScript, Node.js, Nest.js, AWS, EKS, Lambda, PostgreSQL, Redis, Temporal, Terraform, Helm, ArgoCD, Prometheus, Grafana, Jaeger.

Creative IT Park

2021 – 2023

Backend Developer

Islamabad

- Delivered Python/Node.js backend and ETL platforms for university, government, and healthcare systems; improved information discovery with Elasticsearch, Redis, and optimized data access.
- Hardened Kafka pipelines with retries and circuit breakers, secured healthcare data flows with fine-grained AWS controls, and operated Kubernetes deployments with health gating.

GeekGenix – Backend Developer

2020 – 2021

Built Express/MongoDB APIs, Stripe checkout and webhook reconciliation, Passport.js authentication, and Jest-tested customer workflows.

SELECTED AI & KNOWLEDGE-GRAPH PROJECTS

Verifiable Browser Agent Runtime

Python, LangGraph, Graphiti, Neo4j, Chroma, FastAPI

- Architected an asynchronous agent runtime with capture, assessment, policy gating, execution, semantic post-condition verification, and bounded recovery for long-running workflows.
- Integrated Graphiti/Neo4j temporal memory and vector retrieval to persist verified and failed episodes and reuse grounded evidence across sessions.

Remote Proctoring & Collusion Detection

Neo4j, OpenVINO, WebRTC

- Built a real-time proctoring system and modeled students, exam takers, and social relationships in Neo4j, updating the graph during exams to surface potentially collusive patterns and time-mapped incidents.

EDUCATION & LANGUAGES

BS Computer Science — PMAS Arid Agriculture University, Rawalpindi

Languages: Urdu (native), English (professional)